

ENGR 213: Ordinary Differential Equations

Lecture 3 (September 16, 2026)

Expanded and Detailed Notes

First-Order Differential Equations: Separable & Linear Methods

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Topics Covered:

- Separable Equations (Method and Examples)
- Linear Equations (Standard Form and Solution Methods)
- Integrating Factors and U-Substitution
- Piecewise-Linear Differential Equations

Part 1: Separable Equations

Definition

A first-order differential equation of the form is called **separable** or said to have **separable variables**:

$$dy/dx = g(x) \cdot h(y)$$

The key property: **The right side can be written as a product of a function of x alone and a function of y alone.**

Examples of Separable Equations:

1. $dy/dx = x^2 y^4 e^{(5x-3y)}$
2. $dy/dx = y + \cos(x)$

Why are these separable?

In both cases, we can rearrange to separate the variables—all y terms on one side, all x terms on the other. The exponential $e^{(5x-3y)} = e^{5x} \cdot e^{-3y}$ allows us to factor the first equation as a product.

Method of Solution (Step-by-Step)

Step 1: Separate the Variables

Start with: $dy/dx = g(x)h(y)$

Rearrange algebraically to get all y's on the left, all x's on the right:

$$p(y) \, dy = g(x) \, dx$$

where $p(y) = 1/h(y)$ (provided $h(y) \neq 0$)

Step 2: Integrate Both Sides

Once separated, integrate both sides with respect to their own variables:

$$\int p(y) \, dy = \int g(x) \, dx$$

This gives:

$$H(y) = G(x) + C$$

where $H(y)$ is an antiderivative of $p(y)$ and $G(x)$ is an antiderivative of $g(x)$, and C is an arbitrary constant.

Step 3: Solve for y (if possible)

The equation $H(y) = G(x) + C$ is the **implicit solution**. If you can solve for y explicitly, you get the **explicit solution** $y = f(x)$.

Example 1: Solve $(1 + x) dy - y dx = 0$

Solution:

Step 1: Rewrite in standard form

Starting equation: $(1 + x) dy - y dx = 0$

Rearrange: $(1 + x) dy = y dx$

Divide both sides by $y(1 + x)$:

$$dy/y = dx/(1 + x)$$

Step 2: Integrate both sides

$$\int dy/y = \int dx/(1 + x)$$

$$\ln|y| = \ln|1 + x| + C$$

Step 3: Solve for y

Exponentiate both sides: $e^{\ln|y|} = e^{\ln|1+x| + C}$

$$|y| = e^{\ln|1+x|} \cdot e^C = |1 + x| \cdot e^C$$

Let $K = \pm e^C$ (arbitrary constant):

$$y = K(1 + x)$$

Example 2: Solve the IVP $dy/dx = -x/y$, $y(4) = -3$

Solution:

Step 1: Separate variables

Starting: $dy/dx = -x/y$

Multiply both sides by y : $y \, dy = -x \, dx$

Step 2: Integrate both sides

$$\int y \, dy = \int -x \, dx$$

$$y^2/2 = -x^2/2 + C$$

Multiply by 2: $y^2 = -x^2 + 2C$

Rearrange: $x^2 + y^2 = C$ (implicit form)

Step 3: Apply the initial condition $y(4) = -3$

Substitute $x = 4$, $y = -3$ into $x^2 + y^2 = C$:

$$(4)^2 + (-3)^2 = C$$

$$16 + 9 = C \rightarrow C = 25$$

Step 4: Write the particular solution

$$x^2 + y^2 = 25$$

This is a circle centered at the origin with radius 5. The initial condition $y(4) = -3$ means we're on the lower half of this circle.

Example 3: Solve $dy/dx = y^2 - 4$

Solution:

Step 1: Separate variables

$$dy/dx = y^2 - 4 = (y - 2)(y + 2)$$

$$\text{Separate: } dy/[(y - 2)(y + 2)] = dx$$

Step 2: Use partial fractions

We decompose $1/[(y - 2)(y + 2)]$:

$$1/[(y - 2)(y + 2)] = A/(y - 2) + B/(y + 2)$$

$$\text{Solving: } 1 = A(y + 2) + B(y - 2)$$

$$\text{When } y = 2: 1 = 4A \rightarrow A = 1/4$$

$$\text{When } y = -2: 1 = -4B \rightarrow B = -1/4$$

$$\text{So: } 1/[(y - 2)(y + 2)] = (1/4)[1/(y - 2) - 1/(y + 2)]$$

Step 3: Integrate both sides

$$(1/4)[1/(y - 2) - 1/(y + 2)] dy = \int dx$$

$$(1/4)[\ln|y - 2| - \ln|y + 2|] = x + C$$

$$(1/4) \ln|(y - 2)/(y + 2)| = x + C$$

$$\ln|(y - 2)/(y + 2)| = 4x + 4C$$

Step 4: Solve for y (implicit form)

$$|(y - 2)/(y + 2)| = e^{4x + 4C} = Ke^{4x} \text{ (where } K = \pm e^{4C}\text{)}$$

Part 2: Linear First-Order Equations

Definition

A first-order differential equation is **linear in y** if it has the form:

$$a_1(x) \, dy/dx + a_0(x) \, y = g(x)$$

where $a_1(x)$, $a_0(x)$, and $g(x)$ are functions of x only.

Standard Form

Divide the equation by $a_1(x)$ to get:

$$dy/dx + P(x)y = f(x)$$

where $P(x) = a_0(x)/a_1(x)$ and $f(x) = g(x)/a_1(x)$

Classification

- **Homogeneous:** When $f(x) = 0$ (right side is zero)
- **Nonhomogeneous:** When $f(x) \neq 0$

General Solution Structure

The general solution has the form:

$$y = y_c + y_p$$

where:

- y_c is the **complementary solution** (solution to the homogeneous equation)
- y_p is a **particular solution** (solution to the nonhomogeneous equation)

Finding the Complementary Solution y_c

Solve the homogeneous equation: $dy/dx + P(x)y = 0$

This is separable! Rearrange:

$$dy/y = -P(x) dx$$

Integrate both sides:

$$\ln|y| = -\int P(x) dx + C$$

$$y_c = Ce^{-\int P(x) dx}$$

The Integrating Factor Method

To find y_p , we use an **integrating factor**:

$$\mu(x) = e^{\int P(x) dx}$$

Multiply the entire standard form equation by $\mu(x)$:

$$e^{\int P(x) dx} \cdot dy/dx + e^{\int P(x) dx} \cdot P(x)y = e^{\int P(x) dx} \cdot f(x)$$

The left side is the derivative of a product:

$$d/dx [e^{\int P(x) dx} \cdot y] = e^{\int P(x) dx} \cdot f(x)$$

Integrate both sides:

$$e^{\int P(x) dx} \cdot y = \int e^{\int P(x) dx} \cdot f(x) dx$$

Solve for y :

$$y = e^{-\int P(x) dx} \cdot \int e^{\int P(x) dx} \cdot f(x) dx$$

Method of Solution (Step-by-Step)

Step 1: Put the equation in standard form $dy/dx + P(x)y = f(x)$

Step 2: Identify $P(x)$

Step 3: Calculate the integrating factor $\mu(x) = e^{\int P(x)dx}$

Step 4: Multiply the standard form by $\mu(x)$

Step 5: Recognize the left side as $d/dx[\mu(x)y]$

Step 6: Integrate both sides and solve for y

Example 4: Solve $dy/dx - 3y = 6$

Solution:

Step 1: Standard form

The equation is already in standard form: $dy/dx + (-3)y = 6$

Step 2: Identify $P(x)$

$P(x) = -3$ (constant function)

Step 3: Calculate the integrating factor

$$\mu(x) = e^{\int (-3)dx} = e^{-3x}$$

Step 4: Multiply the equation by $\mu(x)$

$$e^{-3x} \cdot dy/dx + e^{-3x} \cdot (-3)y = e^{-3x} \cdot 6$$

Step 5: Recognize the derivative

$$d/dx[e^{-3x} \cdot y] = 6e^{-3x}$$

Step 6: Integrate both sides

$$\int d/dx[e^{-3x} \cdot y] dx = \int 6e^{-3x} dx$$

$$e^{-3x} \cdot y = 6 \cdot (-1/3)e^{-3x} + C = -2e^{-3x} + C$$

Divide by e^{-3x} :

$$y = -2 + Ce^{3x}$$

General Solution: $y = -2 + Ce^{3x}$

Example 5: Solve $(x^2 - 9) dy/dx + xy = 0$

Solution:

Step 1: Put in standard form

Divide by $(x^2 - 9)$:

$$dy/dx + [x/(x^2 - 9)]y = 0$$

Step 2: Identify $P(x)$

$$P(x) = x/(x^2 - 9)$$

Step 3: Calculate the integrating factor

$$\mu(x) = e^{\int [x/(x^2-9)]dx}$$

For the integral $\int x/(x^2 - 9) dx$, use u-substitution: $u = x^2 - 9$, $du = 2x dx$

$$\int x/(x^2 - 9) dx = (1/2) \int du/u = (1/2) \ln|u| = (1/2) \ln|x^2 - 9|$$

$$\mu(x) = e^{(1/2) \ln|x^2-9|} = |x^2 - 9|^{1/2} = \sqrt{|x^2 - 9|}$$

Step 4-6: Multiply by $\mu(x)$, integrate

The integration becomes simpler when we multiply through and recognize that the left side becomes a perfect derivative.

Final result (after integration and simplification):

$$y = C(x^2 - 9)^{-1/2}$$

Key Takeaways

1. Separable Equations

- Recognize when $dy/dx = g(x)h(y)$
- Separate: $p(y)dy = g(x)dx$
- Integrate both sides
- Solve for y if possible
- Always check for singular solutions where $h(y) = 0$

2. Linear Equations

- Put in standard form: $dy/dx + P(x)y = f(x)$
- Calculate integrating factor: $\mu(x) = e^{\int P(x)dx}$
- Multiply entire equation by $\mu(x)$
- Recognize left side as $d/dx[\mu(x)y]$
- Integrate and solve for y
- General solution = complementary + particular

3. U-Substitution for Integration

- Choose u = an expression in the integrand
- Find du = derivative dx
- Rewrite the integral in terms of u
- Integrate
- Substitute back to get answer in terms of original variable

4. When to Use Which Method

- **Separable:** When you can split into $g(x)$ and $h(y)$
- **Linear:** When the equation is first-order and y appears linearly
- **U-Substitution:** When integrating composite functions or products